

finalnotebook

May 13, 2026

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GitHub: <https://github.com/kyletam1/sleep-analysis-project> Problem Statement

This project examines how demographic, behavioral, and health-related factors such as age, physical activity, stress level, and BMI affect sleep duration. The objective is to identify which variables are most impactful in order to better understand patterns associated with healthy sleep.

```
[1]: # from google.colab import drive
      # drive.mount('/content/drive')
```

```
[2]: # Import Packages

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn import preprocessing
from sklearn.linear_model import LinearRegression
from sklearn.feature_selection import r_regression
from sklearn.metrics import r2_score, mean_squared_error
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder
from sklearn.model_selection import train_test_split
from sklearn.pipeline import Pipeline
from sklearn.ensemble import RandomForestRegressor
```

```
[3]: # Upload Dataset

SleepHealth = pd.read_csv('ss.csv')
```

```
[4]: # Display Original Dataset

SleepHealth.head()
```

```
[4]:
```

	Person	ID	Gender	Age	Occupation	Sleep Duration	\
0		1	Male	27	Software Engineer	6.1	
1		2	Male	28	Doctor	6.2	

2	3	Male	28	Doctor	6.2
3	4	Male	28	Sales Representative	5.9
4	5	Male	28	Sales Representative	5.9

	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category \
0	6	42	6	Overweight
1	6	60	8	Normal
2	6	60	8	Normal
3	4	30	8	Obese
4	4	30	8	Obese

	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
0	126/83	77	4200	NaN
1	125/80	75	10000	NaN
2	125/80	75	10000	NaN
3	140/90	85	3000	Sleep Apnea
4	140/90	85	3000	Sleep Apnea

```
[5]: # Remove Duplicates
```

```
SleepHealth = SleepHealth.drop(columns=['Person ID'])
SleepHealth = SleepHealth.drop_duplicates()
```

```
[6]: # Encoding Gender
```

```
SleepHealth['Gender'] = SleepHealth['Gender'].map({
    'Female': 0,
    'Male': 1
}).astype(int)
```

```
[7]: # Fix BMI Category
```

```
SleepHealth['BMI Category'] = SleepHealth['BMI Category'].replace({'Normal': 'Normal Weight'})

SleepHealth = pd.get_dummies(
    SleepHealth,
    columns=['BMI Category'],
    drop_first=True
)

SleepHealth['BMI Category_Obese'] = SleepHealth['BMI Category_Obese'].
    .astype(int)
SleepHealth['BMI Category_Overweight'] = SleepHealth['BMI Category_Overweight'].
    .astype(int)
```

```
[8]: # Split Blood Pressure

SleepHealth[['Systolic BP', 'Diastolic BP']] = SleepHealth['Blood Pressure'].
    ↪str.split('/', expand=True)
SleepHealth = SleepHealth.drop(columns=['Blood Pressure'])

SleepHealth['Systolic BP'] = SleepHealth['Systolic BP'].astype(int)
SleepHealth['Diastolic BP'] = SleepHealth['Diastolic BP'].astype(int)
```

```
[9]: # Fix Sleep Disorder

SleepHealth['Sleep Disorder'] = SleepHealth['Sleep Disorder'].fillna('None')

SleepHealth['Sleep Disorder'] = pd.Categorical(
    SleepHealth['Sleep Disorder'],
    categories=['None', 'Insomnia', 'Sleep Apnea']
)

SleepHealth = pd.get_dummies(
    SleepHealth,
    columns=['Sleep Disorder'],
    drop_first=True
)

SleepHealth['Sleep Disorder_Insomnia'] = SleepHealth['Sleep Disorder_Insomnia'].
    ↪astype(int)
SleepHealth['Sleep Disorder_Sleep Apnea'] = SleepHealth['Sleep Disorder_Sleep_
    ↪Apnea'].astype(int)
```

```
[10]: # Display Cleaned Dataset
```

```
SleepHealth.head()
```

```
[10]:
```

	Gender	Age	Occupation	Sleep Duration	Quality of Sleep \
0	1	27	Software Engineer	6.1	6
1	1	28	Doctor	6.2	6
3	1	28	Sales Representative	5.9	4
5	1	28	Software Engineer	5.9	4
6	1	29	Teacher	6.3	6

	Physical Activity Level	Stress Level	Heart Rate	Daily Steps \
0	42	6	77	4200
1	60	8	75	10000
3	30	8	85	3000
5	30	8	85	3000
6	40	7	82	3500

	BMI Category_Obese	BMI Category_Overweight	Systolic BP	Diastolic BP	\
0	0	1	126	83	
1	0	0	125	80	
3	1	0	140	90	
5	1	0	140	90	
6	1	0	140	90	

	Sleep Disorder_Insomnia	Sleep Disorder_Sleep Apnea
0	0	0
1	0	0
3	0	1
5	1	0
6	1	0

Discussion

For the data cleaning and preprocessing stage, duplicate observations were first removed from the dataset to improve data quality and reduce redundancy. Next, categorical variables were converted into numerical representations so they could be used in the regression models. This preprocessing included encoding variables such as gender, BMI category, and sleep disorder status. Additionally, blood pressure values were separated into systolic and diastolic measurements so they could be analyzed as individual numerical predictors.

[11]: *# Variables*

```
print(" #      Column                                Non-Null Count  Dtype  ")
print("----  -")
print("-----")

for i, col in enumerate(SleepHealth.columns):
    non_null = SleepHealth[col].notnull().sum()
    dtype = SleepHealth[col].dtype

    print(f" {i:<3} {col:<27}{str(non_null) + ' non-null':<16}{dtype}")
```

#	Column	Non-Null Count	Dtype
0	Gender	132 non-null	int64
1	Age	132 non-null	int64
2	Occupation	132 non-null	object
3	Sleep Duration	132 non-null	float64
4	Quality of Sleep	132 non-null	int64
5	Physical Activity Level	132 non-null	int64
6	Stress Level	132 non-null	int64
7	Heart Rate	132 non-null	int64
8	Daily Steps	132 non-null	int64
9	BMI Category_Obese	132 non-null	int64
10	BMI Category_Overweight	132 non-null	int64
11	Systolic BP	132 non-null	int64
12	Diastolic BP	132 non-null	int64

```

13 Sleep Disorder_Insomnia      132 non-null    int64
14 Sleep Disorder_Sleep Apnea  132 non-null    int64

```

```
[12]: # Define Predictors
```

```

x = SleepHealth.drop(columns=['Sleep Duration', 'Occupation', 'Quality of Sleep'])
y = SleepHealth['Sleep Duration']

```

```
[13]: # Standardize data
```

```

Continuous = ['Age', 'Physical Activity Level', 'Stress Level', 'Heart Rate', 'Daily Steps', 'Systolic BP', 'Diastolic BP']

Standardize = preprocessing.StandardScaler()
x[Continuous] = Standardize.fit_transform(x[Continuous])

x.head()

```

```

[13]:
  Gender      Age  Physical Activity Level  Stress Level  Heart Rate \
0      1 -1.609111          -0.803973      0.266533    1.195226
1      1 -1.495222           0.078763      1.420054    0.782756
3      1 -1.495222          -1.392463      1.420054    2.845108
5      1 -1.495222          -1.392463      1.420054    2.845108
6      1 -1.381333          -0.902055      0.843293    2.226402

  Daily Steps  BMI Category_Obese  BMI Category_Overweight  Systolic BP \
0    -1.385485                0                1    -0.303188
1     1.910746                0                0    -0.431459
3    -2.067464                1                0     1.492616
5    -2.067464                1                0     1.492616
6    -1.783306                1                0     1.492616

  Diastolic BP  Sleep Disorder_Insomnia  Sleep Disorder_Sleep Apnea
0    -0.255166                0                0
1    -0.752929                0                0
3     0.906280                0                1
5     0.906280                1                0
6     0.906280                1                0

```

```
[14]: # Split Training / Testing
```

```

X = x[['Age', 'Physical Activity Level', 'Stress Level', 'Heart Rate', 'Daily Steps', 'BMI Category_Obese', 'BMI Category_Overweight', 'Systolic BP', 'Diastolic BP', 'Sleep Disorder_Insomnia', 'Sleep Disorder_Sleep Apnea']]
Y = y

```

```
X_train, X_test, Y_train, Y_test = train_test_split(
    X, Y,
    test_size=0.2,
    random_state=42
)
```

[15]: *# Training Model*

```
Model = LinearRegression()
Model.fit(X_train, Y_train)
```

[15]: LinearRegression()

[16]: *# Testing Model*

```
Y_pred = Model.predict(X_test)

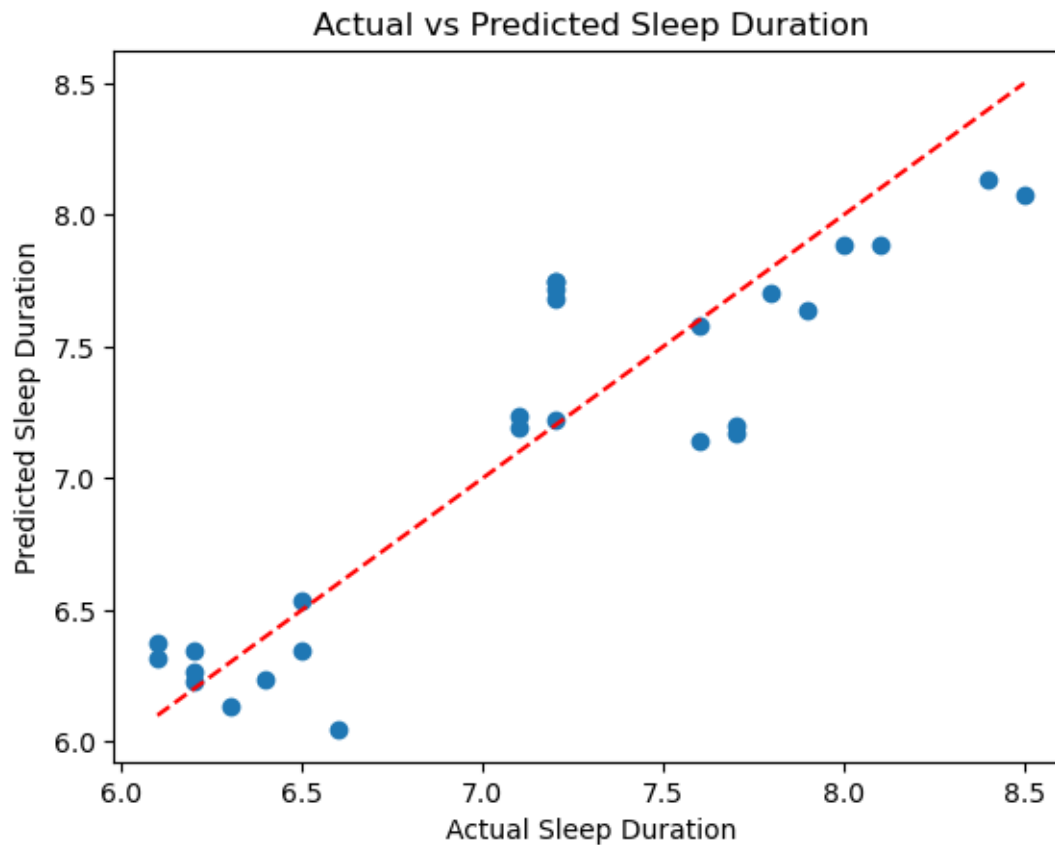
R2 = r2_score(Y_test, Y_pred)
RMSE = np.sqrt(mean_squared_error(Y_test, Y_pred))

print("R²:", R2)
print("RMSE:", RMSE)
```

R²: 0.8055253469955526
RMSE: 0.32160799793951284

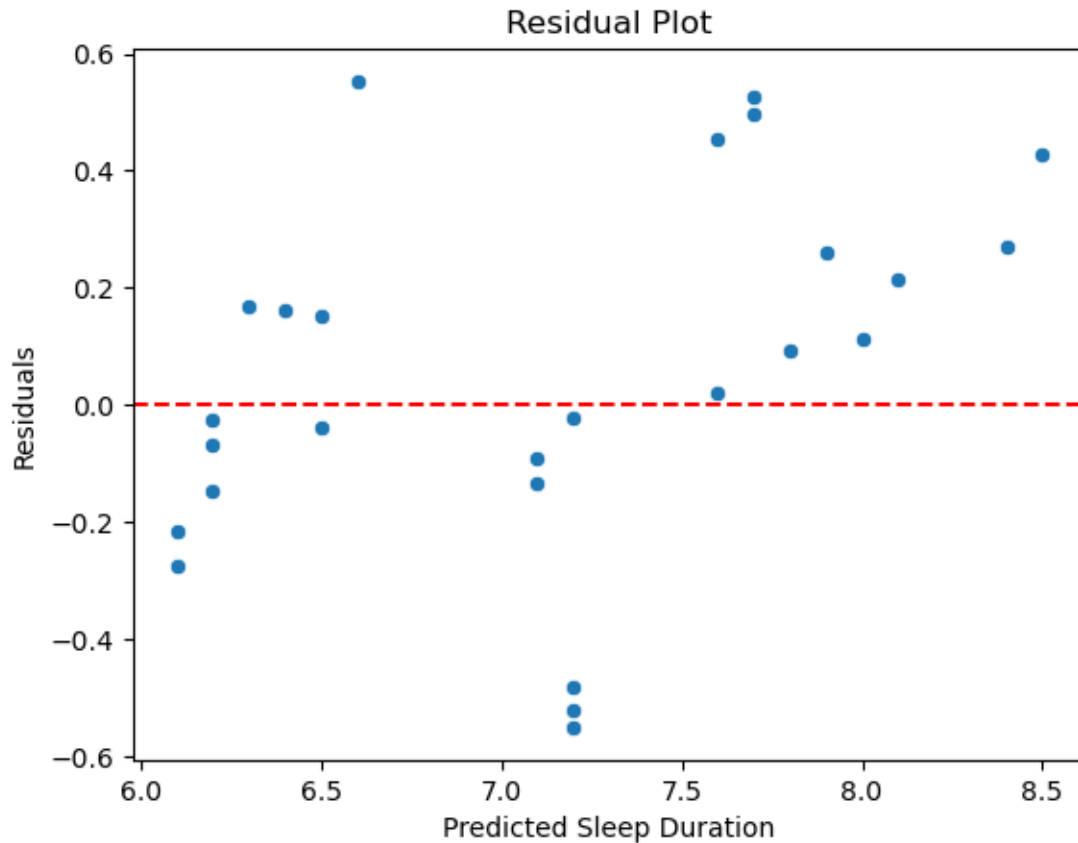
[17]: *# Predicted vs Actual Plot*

```
plt.scatter(Y_test, Y_pred)
plt.xlabel("Actual Sleep Duration")
plt.ylabel("Predicted Sleep Duration")
plt.title("Actual vs Predicted Sleep Duration")
plt.plot([min(Y_test), max(Y_test)], [min(Y_test), max(Y_test)], color='red',
↪linestyle='--')
plt.show()
```



```
[18]: # Residual Plot

Residuals = Y_test - Y_pred
sns.scatterplot(x=Y_test, y=Residuals)
plt.axhline(y=0, color='r', linestyle='--')
plt.xlabel("Predicted Sleep Duration")
plt.ylabel("Residuals")
plt.title("Residual Plot")
plt.show()
```



[19]: *# Coefficient Table*

```
coefficients = pd.DataFrame({
    'Feature': X.columns,
    'Coefficient': Model.coef_.flatten()
})

coefficients = coefficients.reindex(
    coefficients['Coefficient'].abs().sort_values(ascending=False).index
)

print(coefficients)
```

	Feature	Coefficient
6	BMI Category_Overweight	-0.608344
2	Stress Level	-0.384024
7	Systolic BP	-0.304095
1	Physical Activity Level	0.265084
0	Age	0.253485
5	BMI Category_Obese	-0.235471

4	Daily Steps	-0.226913
8	Diastolic BP	0.192482
9	Sleep Disorder_Insomnia	-0.069324
3	Heart Rate	-0.049349
10	Sleep Disorder_Sleep Apnea	0.041736

Discussion

We used a multivariable linear regression model to examine the relationship between various health and lifestyle factors and the amount of sleep an individual gets. The variables included gender, age, physical activity level, stress level, heart rate, daily steps, BMI category, blood pressure, and sleep disorder status.

The model achieved an R^2 value of 0.806 and an RMSE of 0.322. These results suggest that the linear regression model had relatively strong predictive performance and was able to explain a large portion of the variation in sleep duration.

From the predicted versus actual graph and the residual plot, we observed a fairly even spread of points with no strong curves or visible patterns. This suggests that the assumptions of linear regression fit the data reasonably well.

Based on the coefficient table, being classified as overweight had the strongest negative association with sleep duration, while physical activity level had the strongest positive association with sleep duration. This suggests that higher physical activity levels may contribute to longer sleep duration, whereas being overweight may be associated with reduced sleep duration.

```
[20]: # Random Forest Model

Forest = RandomForestRegressor(n_estimators=100, random_state=42)
Forest.fit(X_train, Y_train.values.ravel())
```

```
[20]: RandomForestRegressor(random_state=42)
```

```
[21]: # Testing Model

Y_pred_forest = Forest.predict(X_test)

R2_forest = r2_score(Y_test, Y_pred_forest)
RMSE_forest = np.sqrt(mean_squared_error(Y_test, Y_pred_forest))

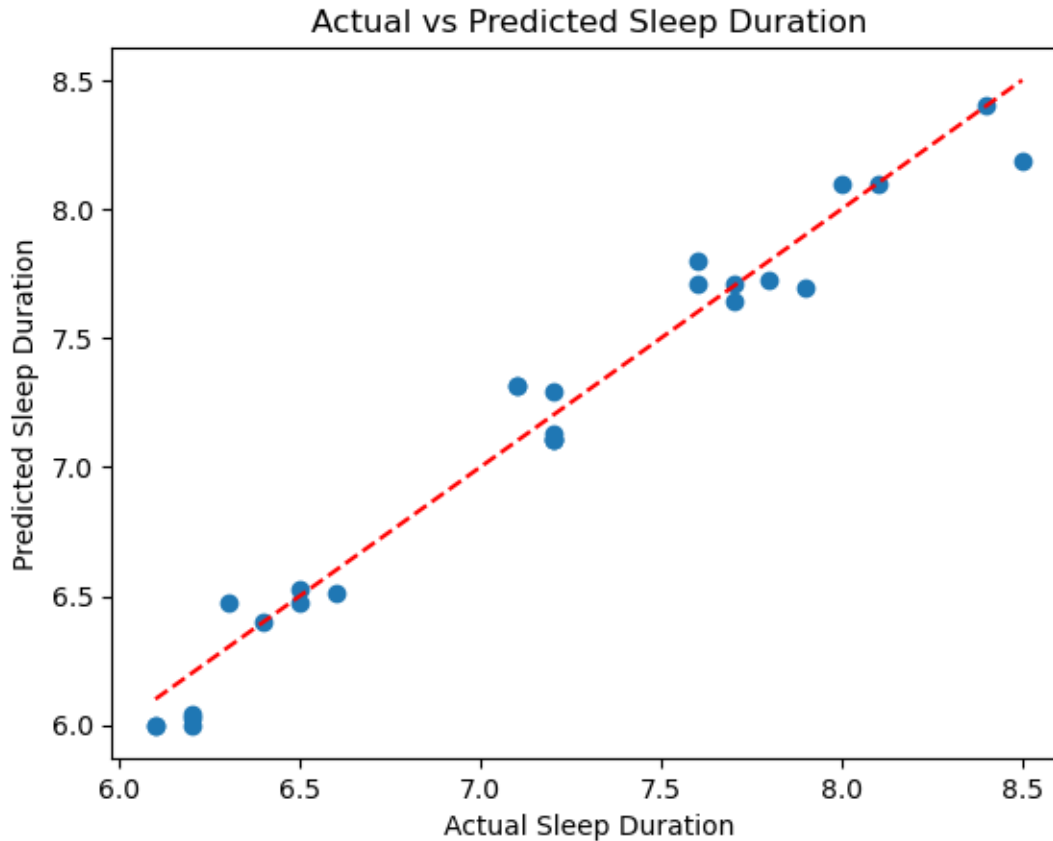
print("R²:", R2_forest)
print("RMSE:", RMSE_forest)
```

```
R²: 0.9654259258253913
RMSE: 0.13560341210983187
```

```
[22]: # Predicted vs Actual Graph

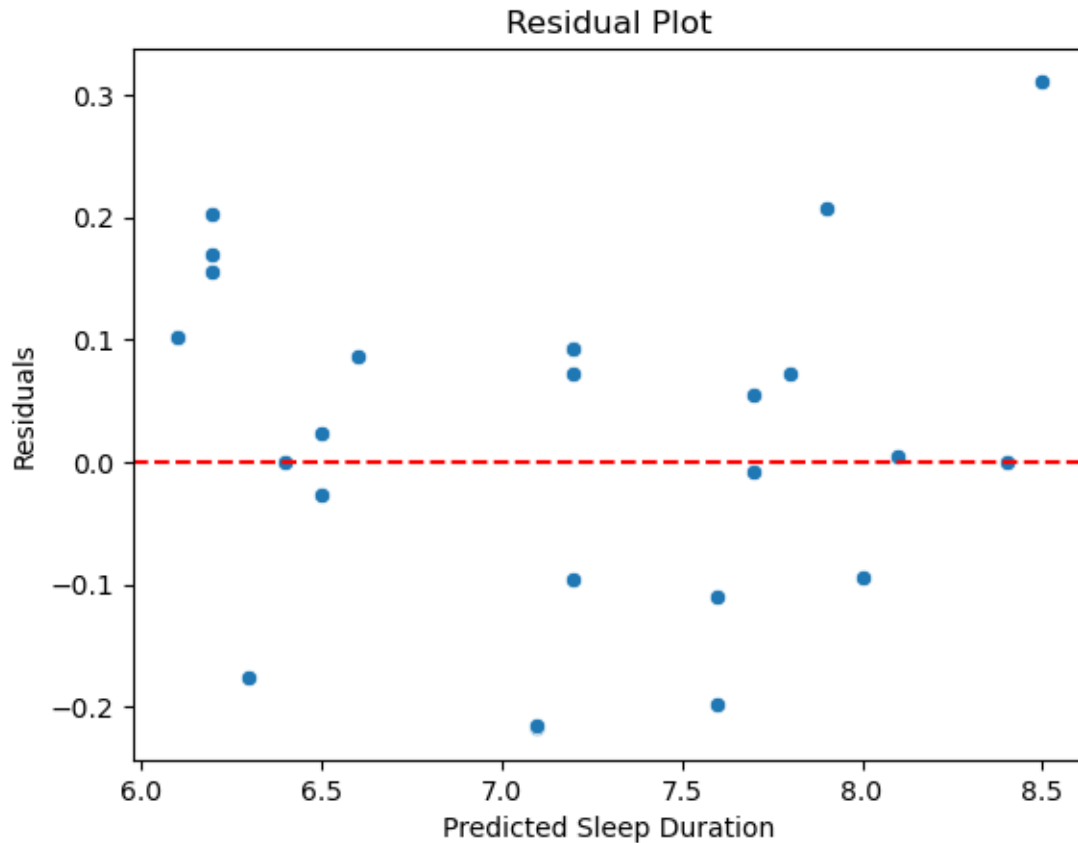
plt.scatter(Y_test, Y_pred_forest)
plt.xlabel("Actual Sleep Duration")
plt.ylabel("Predicted Sleep Duration")
```

```
plt.title("Actual vs Predicted Sleep Duration")
plt.plot([min(Y_test), max(Y_test)], [min(Y_test), max(Y_test)], color='red',
↪linestyle='--')
plt.show()
```



```
[23]: # Residual Plot

Residuals_forest = Y_test - Y_pred_forest
sns.scatterplot(x=Y_test, y=Residuals_forest)
plt.axhline(y=0, color='r', linestyle='--')
plt.xlabel("Predicted Sleep Duration")
plt.ylabel("Residuals")
plt.title("Residual Plot")
plt.show()
```



```
[24]: # Importance Table

importance = pd.DataFrame({
    'Feature': X.columns,
    'Importance': Forest.feature_importances_
})

print(importance.sort_values(by='Importance', ascending=False))
```

	Feature	Importance
2	Stress Level	0.710507
1	Physical Activity Level	0.134947
0	Age	0.081290
6	BMI Category_Overweight	0.031103
3	Heart Rate	0.017172
4	Daily Steps	0.012831
8	Diastolic BP	0.006074
7	Systolic BP	0.004625
10	Sleep Disorder_Sleep Apnea	0.000718
9	Sleep Disorder_Insomnia	0.000627

Discussion

We also implemented a Random Forest regression model to compare its predictive performance with the multivariable linear regression model. A Random Forest model is an ensemble machine learning method that combines many decision trees together to make predictions. Each decision tree is trained on random subsets of the data and predictors, allowing the model to learn different patterns within the dataset. The final prediction is then generated by averaging the predictions from all of the trees in the forest. This process helps improve prediction accuracy and reduce overfitting compared to a single decision tree.

The Random Forest model achieved an R^2 value of 0.965 and an RMSE of 0.136, which was substantially better than the linear regression model. These results suggest that there may be nonlinear relationships and interactions between the predictor variables and sleep duration that were not fully captured by the linear regression model.

The actual versus predicted graph for the Random Forest model appeared similar to the linear regression model but showed points clustered more tightly around the reference line, indicating stronger predictive accuracy.

Feature importance analysis was used to determine how much each variable contributed to the predictive performance of the model. Stress level was by far the most influential predictor, followed by physical activity level and age. In contrast, several health-related variables such as blood pressure and sleep disorder status showed relatively low importance values, suggesting they contributed less additional predictive information once the stronger predictors were already considered by the model.

Overall, the Random Forest model demonstrated significantly stronger predictive performance than the linear regression model and provided additional insight into which variables were most influential in predicting sleep duration.

Conclusion

Overall, this project successfully examined how demographic, behavioral, and health-related factors are associated with sleep duration. After cleaning and preprocessing the dataset, both a multivariable linear regression model and a Random Forest regression model were developed to predict sleep duration using variables such as age, physical activity level, stress level, BMI category, blood pressure, heart rate, daily steps, and sleep disorder status.

The multivariable linear regression model produced an R^2 value of 0.806 and an RMSE of 0.322, indicating that the model was able to explain a large portion of the variation in sleep duration. The residual plots and predicted versus actual graphs suggested that the linear model fit the data reasonably well. Coefficient analysis showed that being classified as overweight and having higher stress levels were associated with shorter sleep duration, while higher physical activity levels and older age were associated with longer sleep duration.

The Random Forest regression model performed even better, achieving an R^2 value of 0.965 and an RMSE of 0.136. This improvement suggests that the relationships between the predictors and sleep duration may contain nonlinear patterns and interactions that were not fully captured by the linear regression model. Feature importance analysis from the Random Forest model revealed that stress level was the most influential predictor by a large margin, followed by physical activity level and age. Variables such as blood pressure and sleep disorder status contributed relatively little additional predictive value compared to the stronger predictors.

In conclusion, the results suggest that stress level, physical activity, age, and BMI are among the most important factors associated with sleep duration. Higher stress levels and being overweight were generally linked to reduced sleep duration, while increased physical activity was associated with improved sleep outcomes. Additionally, the Random Forest model demonstrated that machine learning approaches may provide stronger predictive performance when modeling complex health-related behaviors such as sleep.

Team Distribution

Kyle: Contributed to the problem formulation, interpretation of results, and development of the presentation. He also helped analyze the findings and connect the results to real-world implications.

Andre: Responsible for data cleaning, preprocessing, and implementing the machine learning models, including linear regression and the Random Forest model. He also created the visualizations and performed the technical analysis of the results.

Both team members collaborated on interpreting results and writing the notebook

Overall, stress level was the most influential factor affecting sleep duration across all models.

Project Sharing Statement: This project can be shared with future classes.